



United International University
Department of Computer Science and Engineering
Course Code: CSE 2215 Course Title: Data Structure and
Algorithms I (B)

Class Test – 1 (Marks:20) | Duration: 30mins

Date: 17/07/2021

1.	Derive the best and worst-case time complexity for the following algorithm and express it in O notation. (Here max and min functions assume their usual meaning.) <pre>1. myFunction(A[]) { 2. n = A.length 3. v = n 4. for i = 1 to n { 5. v = min(A[i], v) 6. } 7. v = max(1, n) 8. for j = 1 to 100 { 9. sum = 0 10. k = 1 11. while k <= v { 12. sum = sum + A[k] 13. k = k*2 14. } 15. } 16. }</pre> Hints: Find out the greatest and smallest possible value for v, and use it in your analysis.	[10]
2.	i) Produce an describe an example of an unsuccessful search of Binary search using the values given below after your student id. (You may need to preprocess the numbers to apply binary search, but you do not need to show that preprocessing step)	[5]
	ii) Using the same values produce an describe an example of a successful search of Binary search that takes the maximum possible number of attempts.	[5]

Student ID	Values
011173044	[17, 25, 94, 80, 34, 11, 84, 86, 38, 49]
011181265	[78, 75, 84, 86, 15, 50, 12, 17, 57, 96]
011183017	[77, 68, 67, 93, 71, 95, 10, 59, 74, 61]
011183020	[72, 60, 89, 22, 57, 24, 15, 88, 70, 59]
011183056	[85, 58, 60, 34, 98, 59, 67, 41, 77, 71]

011183077	[14, 83, 82, 66, 25, 27, 76, 10, 61, 62]
011191160	[14, 48, 79, 57, 19, 50, 77, 45, 18, 44]
011191173	[62, 55, 24, 94, 78, 47, 37, 89, 88, 20]
011191179	[65, 44, 93, 91, 97, 13, 32, 77, 51, 24]
011191212	[25, 53, 66, 39, 54, 13, 49, 50, 80, 69]
011192083	[81, 85, 77, 29, 20, 24, 82, 18, 83, 25]
011192094	[74, 52, 85, 27, 71, 81, 35, 97, 98, 31]
011192122	[11, 10, 61, 42, 69, 67, 23, 47, 12, 39]
011193028	[59, 24, 78, 91, 60, 42, 35, 90, 37, 69]
011193038	[66, 16, 65, 37, 63, 91, 42, 67, 74, 32]
011193040	[96, 26, 45, 33, 64, 53, 32, 55, 52, 71]
011193045	[27, 75, 91, 10, 61, 83, 39, 25, 70, 49]
011193047	[19, 70, 85, 44, 36, 50, 13, 14, 97, 57]
011193052	[75, 92, 61, 65, 13, 79, 58, 42, 44, 85]
011193068	[41, 33, 81, 98, 86, 48, 11, 31, 69, 65]
011193122	[60, 51, 12, 62, 81, 54, 88, 27, 80, 48]
011201019	[66, 18, 65, 40, 98, 27, 58, 57, 89, 12]
011201031	[59, 40, 38, 22, 24, 61, 85, 91, 14, 37]
011201037	[26, 19, 39, 55, 49, 30, 75, 27, 54, 63]
011201084	[69, 58, 62, 87, 90, 25, 12, 92, 15, 19]
011201088	[28, 94, 47, 63, 86, 45, 14, 76, 12, 56]
011201108	[85, 46, 70, 23, 14, 66, 86, 72, 44, 62]
011201125	[18, 49, 90, 53, 47, 44, 76, 22, 88, 98]
011201128	[58, 95, 72, 66, 52, 42, 93, 91, 24, 18]
011201156	[63, 92, 49, 78, 94, 39, 69, 22, 80, 86]
011201166	[94, 86, 83, 18, 30, 79, 34, 19, 71, 36]
011201170	[71, 53, 70, 44, 64, 83, 84, 39, 78, 98]
011201173	[12, 69, 41, 46, 58, 59, 22, 80, 87, 96]
011201176	[62, 90, 65, 80, 10, 27, 44, 28, 66, 16]
011201286	[26, 24, 98, 68, 62, 69, 91, 63, 56, 14]
011201334	[92, 36, 48, 40, 94, 62, 20, 93, 64, 90]
011201344	[34, 67, 44, 83, 40, 15, 59, 91, 51, 95]
011201352	[73, 45, 86, 32, 43, 67, 33, 98, 17, 87]

011201353	[51, 89, 67, 36, 41, 35, 10, 78, 75, 91]
011201366	[40, 17, 85, 15, 76, 31, 44, 65, 33, 19]
011201400	[53, 85, 71, 73, 90, 31, 97, 48, 12, 24]
011201403	[80, 46, 63, 76, 13, 92, 35, 74, 53, 34]
011201406	[80, 41, 76, 54, 64, 92, 96, 86, 98, 65]
011201446	[84, 11, 13, 65, 90, 36, 67, 92, 10, 63]
011202199	[63, 94, 97, 38, 37, 65, 46, 33, 47, 92]
Not registered	[29, 27, 74, 30, 33, 75, 48, 84, 49, 62]